SUSHMITHA LAVANURU

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PROFESSIONAL SUMMARY

Technically driven Computer Science undergraduate with hands-on experience in data science, machine learning, and AI-driven application development. Skilled in building intelligent applications, applying machine learning techniques to real-world problems, and developing scalable solutions using modern technologies. Seeking opportunities to apply strong problem-solving and technical skills to build impactful, data-driven systems.

EDUCATION

B.Tech CSE - SR International Institute of Technology, Hyderabad, CGPA: 7.75 — 2023-2027
Class XII - Resonance adventures, Telangana, 96% — 2023
Class X - Narayana Olympiad School, Andhra Pradesh, 99.3% — 2021

PROJECTS

Hospital Patient Readmission Risk Prediction |GitHub

- Built a machine learning classifier to predict 30-day hospital readmission risk on the UCI Diabetes 130-US Hospitals dataset (81,414 patient encounters), targeting a problem linked to Medicare (CMS) readmission penalties.
- Performed **data cleaning and feature engineering**: created 31 model-ready features from 50 raw columns by aggregating 20+ medication columns and mapping ICD diagnosis codes into 8 categories and removed duplicate patient encounters to prevent data leakage.
- Trained and evaluated **Logistic Regression and Decision Tree models** with class-weight balancing for a **9% positive-class imbalance**, optimizing for recall because missed readmissions cost more than false alarms.
- Conducted a subgroup fairness audit by age group that revealed a significant recall gap and diagnosed and fixed 6 data and evaluation bugs.
- Tools & technologies used: Python, pandas, scikit-learn, **Logistic Regression, Decision Tree.**

Content Maker – Autonomous Multi-Agent YouTube Production Engine | GitHub

- Architected a cyclic **multi-agent LLM** pipeline in **LangGraph** with specialized agents (Research, Script Writer, Critic, Script Doctor, SEO, Thumbnail, Repurposing) that converts a topic idea into a publish-ready YouTube package: script, SEO metadata, chapter timestamps, and thumbnail concepts.
- Implemented a **6-dimension LLM-critic quality rubric** (1–5 scale) that scores each draft and automatically routes low-scoring drafts to a separate revision agent, with a **3-loop limit and automated human escalation**.
- Improved rubric score from **2/5 to 5/5** through iterative refinement across 3 test scenarios (LLM-judge scoring).
- Added parallel content packaging (SEO and thumbnail agents), version-diff tracking, Pydantic-validated agent outputs, and isolated test suites for each agent, served through a FastAPI backend with a React dashboard.
- Tools & technologies used: **Python, LangGraph, LLMs (Groq),** Pydantic, FastAPI, React.

IPL Analytics Dashboard | GitHub

- Cleaned, transformed, and modeled 200,000+ rows of IPL data (2008–2023) covering matches, players, venues, toss decisions, and season outcomes.
- Wrote MySQL reporting queries and pre-calculated tournament winner, runner-up, Orange Cap, and Purple Cap metrics to improve dashboard query performance.
- Designed an interactive Power BI dashboard with season slicers, KPI cards, points tables, toss analysis, and venue analysis for non-technical users.
- Tools & technologies used: **MySQL, SQL, Power BI**, Data Cleaning, Data Visualization.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript
AI & ML: Agentic AI, Multi-Agent Systems, LLMs, Prompt Engineering, Groq API
Frameworks and Libraries: scikit-learn, pandas, NumPy, matplotlib, LangGraph, FastAPI, React
Developer Tools and Platforms: Git, GitHub, Jupyter Notebook
Databases: MySQL

LEADERSHIP & ARCHIVEMENTS

Volunteer, Cultural Club

Volunteer and Winner, Debate Club

CERTIFICATIONS

- Data Analytics Job Simulation: Data Analysis & Forensic Technology - Deloitte via Forage
- Data Visualisation: Empowering Business with Effective Insights - TATA via Forage
- Google AI Essentials - Google Career Certificates, Credly Verified
- SQL Essential Training, 7.8 CPE Credits NASBA Accredited - LinkedIn Learning
- Introduction to Data Science – Cisco, Credly Verified